CHAPTER 7 TECHNIQUES OF INTEGRATION

7.1 Integration by Parts

分部积式 ? 常见 第一次一定微后为0 endless

- **Integration by Parts**

$$\int udv = uv - \int vdu$$

$$\int_a^b u(x)v'(x)dx = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x)dx$$

$(uv)' = u\cdot\frac{dv}{dx} + v\frac{du}{dx}$

$= uv' + u'v$

$\int (uv)'dx = \int uv' + u'v\,dx$

$uv = \int uv'dx + \int u'v\,dx$

$\int uv'dx = uv - \int u'v\,dx$

let $dv = v'dx$; $du = u'dx$

$\int u\,dv = uv - \int v\,du$

Example 7.1. Find $\int x \cos x dx$. $\Rightarrow \int u dv = uv - \int v du$

let $u = x$ $dv = \cos x dx$

$\frac{du}{dx} = 1$ $v = \sin x$

$du = dx$

$\int x \cos x dx = x \sin x - \int \sin x dx$

$= x \sin x + \cos x + C$

Selection for u:

priority

Logarithmic

Inverse trigo

Algebraic (x^n)

Trigo

Exponential (a^x)

special case (product but use the rule for product)

Example 7.2. Find $\int_1^2 \ln x dx$.

$= \int_1^2 \ln x \cdot 1 \, dx$

$\int u dv = uv - \int v du$

let $u = \ln x$, $dv = dx$

$du = \frac{1}{x} dx$ $v = x$

$= [\ln x \cdot x]_1^2 - \int_1^2 x \cdot \frac{1}{x} dx$

$= [x \ln x - x]_1^2 + C$

$= 2 \ln 2 - 1$

$\int_1^2 \ln x \, dx = x \ln x - \frac{x}{2}$

LIATE

Example 7.3. Find $\int \sin^{-1} x\,dx$.

Inver trig.

$(\sin^{-1}x)' = \frac{1}{\sqrt{1-x^2}}$

let $u = \sin^{-1}x$; $dv = dx$

$du = \frac{1}{\sqrt{1-x^2}}dx$, $v = x$

$\int u\,dv = uv - \int v\,du$

$\int \sin^{-1}x\,dx = \sin^{-1}x \cdot x - \int x \cdot \frac{1}{\sqrt{1-x^2}}dx$

let $u = 1-x^2$; $\frac{du}{dx} = -2x$

$dx = \frac{du}{-2x}$

$= x\sin^{-1}x - \int x \cdot \frac{1}{\sqrt{u}} \cdot \frac{1}{-2x}\,du$

$= x\sin^{-1}x + \frac{1}{2}\int \frac{1}{\sqrt{u}}\,du$

$= x\sin^{-1}x + u^{\frac{1}{2}} + C$

check

Example 7.4. Find $\int_1^2 t^6 \ln t\,dt$.

LIATE

let $u = \ln t$; $dv = t^6 dt$

$\frac{du}{dt} = \frac{1}{t}$; $v = \frac{t^7}{7}$

$\int_1^2 t^6 \ln t\,dt = \ln t \cdot \left(\frac{t^7}{7}\right) - \frac{1}{7}\int t^7 \cdot \frac{1}{t}\,dt$

$= \frac{1}{7}\cdot t^7 \ln t - \frac{1}{49}\cdot t^7$

LIATE

Example 7.5. Find $\int x^2 \sin x dx$.

let $u = x^2$, $dv = \sin x\,dx$

$du = 2x\,dx$, $v = -\cos x$

$$\int x^2 \sin x dx = x^2 \cdot (-\cos x) - \int -\cos x \cdot (2x)\,dx$$

composite rule not applicable

$$= -x^2 \cos x + 2\int x \cos x\,dx$$

let $u = x$, $dv = \cos x\,dx$

$du = dx$, $v = \sin x$

$$= -x^2 \cos x + 2(x \cdot \sin x - \int \sin x\,dx)$$

$$= -x^2 \cos x + 2x \sin x + 2\cos x + C$$

LIATE

Example 7.6. Find $\int e^x \sin x dx$.

let $u = \sin x$, $dv = e^x \sin x\,dx$

$du = \cos x\,dx$, $v = -\cos x\, e^x$

$$\int e^x \sin x dx = \sin x \cdot (-\cos x) - \int -\cos x \cdot \cos x\,dx$$

let $u = \cos x$, $dv = e^x dx$

$du = -\sin x\,dx$, $v = e^x$

$$= -e^x \sin x - \int e^x \cos x\,dx$$

$$= -e^x \sin x - (e^x \cos x + \int e^x \sin x\,dx)$$

$$= -e^x \sin x - e^x \cos x - \int e^x \sin x\,dx$$

eg $x = 5 - 4 - x$

solve for x

$$2\int e^x \sin x\,dx = -e^x \sin x - e^x \cos x + C_1$$

$$\int e^x \sin x\,dx = \frac{1}{2} e^x (-\sin x - \cos x) + C_2$$

$\int x^5 \sin x \, dx$

$= -x^5\cos x - (-5x^4\sin x) + (20x^3\cos x)$

$- 60x^2\sin x + (-120x\cos x) - (-120\sin x)$

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let $u = x^5$; $du = 5x^4\,dx$

$dv = \sin x\,dx$; $v = -\cos x$

$= -x^5\cos x - \int -\cos x \cdot 5x^4\,dx$

$= -x^5\cos x + 5\int \cos x \cdot x^4\,dx$

let $u = x^4$; $du = 4x^3\,dx$

$dv = \cos x\,dx$; $v = \sin x$

$= -x^5\cos x + 5(x^4\sin x - \int \sin x \cdot 4x^3\,dx)$

$= -x^5\cos x + 5x^4\sin x - 20\int x^3\sin x\,dx$

当 u 是 fun 为 0 的为止

D(ifferation) I(nteg...)

x^5	$\sin x$
$5x^4$	$-\cos x$
$20x^3$	$-\sin x$
$60x^2$	$\cos x$
$120x$	$\sin x$
120	$-\cos x$
0	$-\sin x$

$\int u\,dv = u_1v_1 - \int v_1\,du_1$

$= u_1v_1 - [u_2v_2 - \int v_2\,du_2]$

$= u_1v_1 - u_2v_2 + \int v_2\,du_2$

$= u_1v_1 - u_2v_2 + u_3v_3 - \int v_3\,du_3$

Δ Δ Δ Δ

plus minus plus minus

Not important

Example 7.7. Derive a reduction formula for $\int \sin^n xdx$. Hence, evaluate $\int_0^{\frac{\pi}{2}} \sin^8 xdx$.

Exercises 7.1 3, 5, 7, 9, 11, 13, 15, 17, 19, 23, 27, 29, 33, 35, 37, 41, 47, 49, 51, 53, 61, 63, 65.

7.2 Some Trigonometric Integrals

Odd power : add → even

Example 7.8. Find $\int \sin^5 x dx$.

remove odd power

$\sin^2\theta + \cos^2\theta = 1$

$$\int \sin^5 x\,dx = \int \sin^4 x \cdot \sin x\,dx$$

$$= \int (\sin^2 x)^2 \sin x\,dx$$

need to be removed

$$= \int (1-\cos^2\theta)^2 \sin x\,dx$$

Let $u = \cos x$; $\frac{du}{dx} = -\sin x$

$dx = \frac{du}{-\sin x}$

$$= \int (1-u^2)^2 \cdot \sin x \cdot \frac{du}{-\sin x}$$

$$= -\int 1 - 2u^2 + u^4\,du$$

$$= -\left(u - \tfrac{2}{3}u^3 + \tfrac{1}{5}u^5\right) + C$$

$$= -u + \tfrac{2}{3}u^3 - \tfrac{1}{5}u^5 + C$$

Even power ? reduce to power 1

Example 7.9. Find $\int \cos^4 x dx$.

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= 2\cos^2\theta - 1 \Rightarrow \cos^2\theta = \frac{\cos 2\theta + 1}{2}$

$= 1 - 2\sin^2\theta$

$$= \int (\cos^2 x)^2\,dx$$

$$= \int \left(\frac{\cos 2x + 1}{2}\right)^2 dx$$

$$= \frac{1}{4}\int (\cos 2x + 1)^2\,dx$$

$\frac{\cos 4x + 1}{2}$

$$= \frac{1}{4}\int \cos^2 2x + 2\cos 2x + 1\,dx$$

$\cos^2 2x + 2\cos 2x + 1$

$\sin^3 2x$

$$= \frac{1}{4} \times \frac{1}{6}\sin^3 2x + \frac{1}{4}\sin 2x + \frac{1}{4}x + C$$

$$\frac{1}{4} \times \frac{1}{2} \times \frac{1}{4}\sin 4x + \frac{1}{4}\frac{x}{2}$$

TRY

Example 7.10. Find $\int \frac{\sin^3 x}{\cos^4 x} dx$. (odd / even)

let $u = \cos x$; $du = -\sin x\, dx \Rightarrow dx = \frac{du}{-\sin x}$

$$= \int \frac{\sin^2 x \cdot \sin x}{\cos^4 x} dx = \int \frac{(1-\cos^2 x)\sin x}{\cos^4 x} dx$$

$$= \int \frac{1-u^2 \cdot \sin x}{u^4} \cdot \frac{du}{-\sin x}$$

$$= -\int \frac{1-u^2}{u^4} du$$

$$= -\int u^{-4} - u^{-2}\, du$$

$$= -\left(\frac{u^{-3}}{-3} + u^{-1}\right) + C = +\frac{u^{-3}}{3} - u^{-1} + C$$

$$= \frac{(\cos x)^{-3}}{3} - (\cos x)^{-1} + C$$

$$= \frac{1}{3\cos^3 x} - \frac{1}{\cos x} + C$$

Example 7.11. Find $\int \sin^2 x \cos^4 x dx$.

$$= \int \frac{1-\cos 2x}{2} \cdot (\cos^2 x)^2\, dx$$

$$= \frac{1}{2}\int (1-\cos 2x)\left(\frac{1+\cos 2x}{2}\right)^2 dx$$

$$= \frac{1}{8}\int (1-\cos 2x)(1+2\cos 2x+\cos^2 2x)\, dx$$

$$= \frac{1}{8}\int 1+2\cos 2x+\cos^2 2x-\cos 2x-2\cos^2 2x-\cos^3 2x\, dx$$

$$= \frac{1}{8}\int 1+\cos 2x-\cos^2 2x-\cos^3 2x\, dx$$

$\cos^2 2x \cdot \cos 2x$
$(1-\sin^2 2x)$

$$= \frac{1}{8}\left(x+\frac{\sin 2x}{2}\right) - \frac{1}{8}\int \frac{1+\cos 4x}{2} + (1-\sin^2 2x)\cos 2x\, dx$$

$$= \frac{x}{8} + \frac{\sin 2x}{16} - \frac{1}{16}\int 1+\cos 4x\, dx - \frac{1}{8}\int (1-\sin^2 2x)\cos 2x\, dx$$

let $u = \sin 2x$; $\frac{du}{dx} = 2\cos 2x \Rightarrow dx = \frac{du}{2\cos 2x}$

$$= \frac{x}{8} + \frac{\sin 2x}{16} - \frac{1}{16}x - \frac{1}{64}\sin 4x - \frac{1}{8}\int (1-u^2)\cdot \cos 2x \cdot \frac{du}{2\cos 2x}$$

$$- \frac{1}{16}\int 1-u^2\, du$$

$$- \frac{1}{16}\left(u - \frac{u^3}{3}\right) + C$$

$$- \frac{1}{16}\left(\sin 2x - \frac{\sin^3 2x}{3}\right) + C$$

$$= +\frac{x}{16} + \frac{\sin^2 x}{48} - \frac{1}{64}\sin 4x + C$$

$\cos 2x = \cos^2 x - \sin^2 x$
$2\cos^2 x - 1$
$\cos^2 x = \frac{1+\cos 2x}{2}$

- **Some Trigonometric Identities**

$\sin(a+b) = \sin a\cos b + \cos a\sin b$

$\sin(a-b) = \sin a\cos b - \cos a\sin b$

$2\sin a\cos b = \sin(a+b) + \sin(a-b)$

$\cos(a+b) = \cos a\cos b + \sin a\sin b$

$\cos(a-b) = \cos a\cos b - \sin a\sin b$

$2\cos a\cos b = \cos(a+b) + \cos(a-b)$

$2\sin a\sin b = \cos(a+b) - \cos(a-b)$

(a) $\sin mx\cos nx = \frac{1}{2}[\sin(m+n)x + \sin(m-n)x]$

(b) $\sin mx\sin nx = -\frac{1}{2}[\cos(m+n)x - \cos(m-n)x]$

(c) $\cos mx\cos nx = \frac{1}{2}[\cos(m+n)x + \cos(m-n)x]$

Example 7.12. Find $\int \sin 2x\cos 3x\,dx$.

$= \frac{1}{2}\int[\sin 5x + \sin(-x)]\,dx$ $\quad \sin(-\theta) = -\sin\theta$

$= \frac{1}{2}\int \sin 5x - \sin x\,dx$

$= \frac{1}{2}\left(\frac{-\cos 5x}{5} + \cos x\right) + C$

$= \frac{1}{2}\cos x - \frac{1}{10}\cos 5x + C$

Example 7.13. If m and n are positive integers, show that

$$\int_{-\pi}^{\pi} \sin mx \sin nx dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

if $m \neq n$ (b) $\sin mx \sin nx = -\frac{1}{2}[\cos(m+n)x - \cos(m-n)x]$

$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = -\frac{1}{2}\int_{-\pi}^{\pi} \cos(m+n)x - \cos(m-n)x \, dx$

$= -\frac{1}{2}\left[\frac{-\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n}\right]_{-\pi}^{\pi}$

$= -\frac{1}{2}[0-0] = 0$ ∵ if $m+n$ or $m-n$ are integer, $\sin x\pi = 0$ (integer)

Example 7.14. If m and n are positive integers, find

$$\int_{-L}^{L} \sin\frac{\pi mx}{L} \sin\frac{\pi nx}{L} dx$$

Example 7.13. If m and n are positive integers, show that

$$\int_{-\pi}^{\pi} \sin mx \sin nx dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

$1 - 2\sin^2 x = \cos 2x$

$\sin^2 x = \frac{1 - \cos 2x}{2}$

(b) $\sin mx \sin nx = -\frac{1}{2}[\cos(m+n)x - \cos(m-n)x]$

if $m = n$

$\int_{-\pi}^{\pi} \sin mx \sin nx dx = \int_{-\pi}^{\pi} \sin mx \sin mx dx$

$= \int_{-\pi}^{\pi} \sin^2(mx)\, dx$

$= \int_{-\pi}^{\pi} \frac{1 - \cos(2mx)}{2} dx = \frac{1}{2}\int_{-\pi}^{\pi} 1 - \cos(2mx)\, dx$

$= \frac{1}{2} \cdot [x]_{-\pi}^{\pi} - \frac{1}{2} \cdot \left[\frac{\sin 2mx}{2m}\right]_{-\pi}^{\pi}$

$= \frac{1}{2}[\pi - (-\pi)] = \pi$

#Try

Example 7.14. If m and n are positive integers, find

$$\int_{-L}^{L} \sin \frac{\pi mx}{L} \sin \frac{\pi nx}{L} dx$$

$\begin{cases} 0, & \text{if } m \neq n \\ 1, & \text{if } m = n \end{cases}$

Example 7.15. Find $\int \tan^5 x dx$.

$= \int \tan^3 x \tan^2 x \, dx$

$= \int \tan^3 x (\sec^2 x - 1) \, dx$

$= \int \sec^2 x \tan^3 x - \tan^3 x \, dx$

$= \int \sec^2 x \tan^3 x - (\sec^2 x - 1)\tan x \, dx$

$= \int \sec^2 x \tan^3 x - \sec^2 x \tan x + \tan x \, dx$ ← 分开做一下 logical

let $u = \tan x$ let $u = \tan x$ let $u = \cos x$ refer to the technique

$= \frac{\tan^4 x}{4} - \frac{\tan^2 x}{2} - \ln|\sec x| + C$

$\sin^2\theta + \cos^2\theta = 1$

$\tan^2\theta + 1 = \frac{1}{\cos^2\theta} = \sec^2\theta$

$\tan^2\theta = \sec^2\theta - 1$

$\cot^2\theta = \csc^2\theta - 1$

$\tan^2\theta = \frac{1}{\cos^2\theta} - 1$

$1 + 2\cot x + \cot^2 x$

Example 7.16. Find $\int \cot^4 x dx$.

$\int (\cot^2 x)^2 \, dx$

$= \int (\cos^2 x - 1)^2 \, dx$

$= \int \cos^4 x - 2\cos^2 x - 1 \, dx$

$= \int \left(\frac{\cos 2x + 1}{2}\right)^2 - 2 \times \frac{\cos 2x + 1}{2} - 1 \, dx$

$= \int \frac{\cos^2 2x + 2\cos 2x + 1}{4} - \cos 2x - 2 \, dx$

$= 4\int \frac{\cos 2x + 1}{2} + 2\cos 2x + 1 \, dx - \int \cos 2x \, dx - 2x + C$

$= 4\int \frac{1}{2}\cos 2x + \frac{3}{2} \, dx - \frac{\sin 2x}{2} - 2x + C$

$= 10 \times \frac{\sin 2x}{2} + \frac{3}{2}x - \frac{\sin 2x}{2} - 2x + C$

$= \frac{3}{2}\sin 2x - \frac{x}{2} + C$

$\sin^2 x + \cos^2 x = 1$

$1 + \cot^2 x = \frac{1}{\sin^2 x} = \csc^2 x$

$\tan^2 x + 1 = \sec^2 x$

$\cot^2 x \cdot \cot^2 x$

$\int (\csc^2 x - 1)\cot^2 x \, dx$

$= \int \csc^2 x \cdot \cot^2 x - \cot^2 x \, dx$

$= \int \csc^2 x \cot^2 x - (\csc^2 x - 1) \, dx$

$= \int \csc^2 x \cot^2 x \, dx - \int \csc^2 x - 1 \, dx$

Let $u = \cot x$; $\frac{du}{dx} = -\csc^2 x \Rightarrow dx = \frac{du}{-\csc^2 x}$

$= -\int u^2 \cdot du + \cot x - x + C$

$= -\frac{\cot^3 x}{3} + \cot x - x + C$

$\tan x$ $\sec^2 x$

$\sec x$ $\sec x \tan x$

$\cot x$ $-\csc^2 x$

$\csc x$ $-\csc x \cot x$

$\tan x \quad \sec^2 x$

$\sec x \quad \sec x \tan x$

$\cot x \quad -\csc^2 x$

$\csc x \quad -\csc x \cot x$

Example 7.17. Find $\int \sec x \tan^3 x dx$.

$= \int \sec x \tan^2 x \tan x \, dx$

$= \int \sec x \tan x \tan^2 x \, dx$

$= \int \sec x \tan x (\sec^2 x - 1) dx$

$\sec x \tan x \cdot \tan^2 x$

$\sec x \tan x (\sec^2 x - 1)$

$\cos^2 x + \sin^2 x = 1$

$1 + \tan^2 x = \frac{1}{\cos^2 x} = \sec^2 x$

$\tan^2 x = \sec^2 x - 1$

$\sec^3 x + \tan x - \sec x \tan x$

Let $u = \sec x$; $\frac{du}{dx} = \sec x \tan x \Rightarrow dx = \frac{du}{\sec x \tan x}$

$= \int u^2 - 1 \, du = \frac{u^3}{3} - u$

$= \frac{\sec^3 x}{3} - \sec x$

Example 7.18. Find $\int \tan^6 x \sec^4 x dx$.

$\Rightarrow$ 不能用 substitution，除非有 $\sec^2 x$

使用 $u = \tan x \Rightarrow \frac{du}{dx} = \sec^2 x$

$dx = \frac{du}{\sec^2 x}$

$= \int \tan^6 x \sec^2 x \cdot \sec^2 x \, dx$

$= \int \tan^6 x (\tan^2 x + 1) \sec^2 x \, dx$

$= \int u^6 (u^2 + 1) \cdot \sec^2 x \cdot \frac{du}{\sec^2 x}$

$= \int u^6 (u^2 + 1) \, du = \int u^8 + u^6 \, du = \frac{u^9}{9} + \frac{u^7}{7} + C$

$= \frac{\tan^9 x}{9} + \frac{\tan^7 x}{7} + C$

Strategy for Evaluating $\int \sin^m x \cos^n x\, dx$

(a) If the power of cosine is odd ($n = 2k + 1$), save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of sine:

$$\int \sin^m x \cos^{2k+1} x\, dx = \int \sin^m x\, (\cos^2 x)^k \cos x\, dx$$
$$= \int \sin^m x\, (1 - \sin^2 x)^k \cos x\, dx$$

Then substitute $u = \sin x$.

(b) If the power of sine is odd ($m = 2k + 1$), save one sine factor and use $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of cosine:

$$\int \sin^{2k+1} x \cos^n x\, dx = \int (\sin^2 x)^k \cos^n x \sin x\, dx$$
$$= \int (1 - \cos^2 x)^k \cos^n x \sin x\, dx$$

Then substitute $u = \cos x$. [Note that if the powers of both sine and cosine are odd, either (a) or (b) can be used.]

(c) If the powers of both sine and cosine are even, use the half-angle identities

$$\sin^2 x = \tfrac{1}{2}(1 - \cos 2x) \qquad \cos^2 x = \tfrac{1}{2}(1 + \cos 2x)$$

It is sometimes helpful to use the identity

$$\sin x \cos x = \tfrac{1}{2} \sin 2x$$

Strategy for Evaluating $\int \tan^m x \sec^n x\, dx$

(a) If the power of secant is even ($n = 2k$, $k \geqslant 2$), save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$:

$$\int \tan^m x \sec^{2k} x\, dx = \int \tan^m x\, (\sec^2 x)^{k-1} \sec^2 x\, dx$$
$$= \int \tan^m x\, (1 + \tan^2 x)^{k-1} \sec^2 x\, dx$$

Then substitute $u = \tan x$.

(b) If the power of tangent is odd ($m = 2k + 1$), save a factor of $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to express the remaining factors in terms of $\sec x$:

$$\int \tan^{2k+1} x \sec^n x\, dx = \int (\tan^2 x)^k \sec^{n-1} x \sec x \tan x\, dx$$
$$= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x\, dx$$

Then substitute $u = \sec x$.

Exercises 7.2 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29 (another answer: $\frac{1}{8}\sec^8 x - \frac{1}{6}\sec^6 x + C$), 33, 39, 41, 45, 47, 61, 63.

7.3 Trigonometric Substitutions

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Expression	Substitution	Domain	Identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$0 \le \theta \le \pi,\ \theta \ne \frac{\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

Example 7.19. Find $\int \sqrt{a^2 - x^2}\,dx$.

substitution
integration by parts
trigo

let $x = a\sin\theta$

$\frac{dx}{d\theta} = a\cos\theta$

$dx = a\cos\theta\, d\theta$

$= \int \sqrt{a^2 - a^2\sin^2\theta}\, dx$

$= \int \sqrt{a^2\cos^2\theta}\,(a\cos\theta)\, d\theta$

$= \int a\cos\theta\,(a\cos\theta)\, d\theta$

$= a^2 \int \cos^2\theta\, d\theta$

$\cos 2A = \cos^2 A - \sin^2 A$

$= 2\cos^2 A - 1$

$= a^2 \int \frac{1+\cos 2\theta}{2}\, d\theta$

$= \frac{a^2}{2} \int 1 + \cos 2\theta\, d\theta$

$= \frac{a^2}{2}\left[\theta + \frac{\sin 2\theta}{2}\right] + C \quad = \frac{a^2}{2}$

7.3 Trigonometric Substitutions

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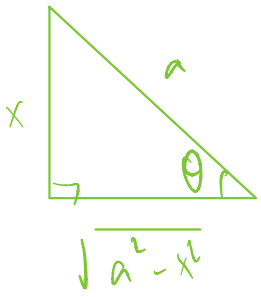
Expression	Substitution	Domain	Identity
$\sqrt{a^2-x^2}$	$x = a\sin\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1-\sin^2\theta = \cos^2\theta$
$\sqrt{a^2+x^2}$	$x = a\tan\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1+\tan^2\theta = \sec^2\theta$
$\sqrt{x^2-a^2}$	$x = a\sec\theta$	$0 \le \theta \le \pi,\ \theta \ne \frac{\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

Example 7.19. Find $\int \sqrt{a^2-x^2}\,dx$.

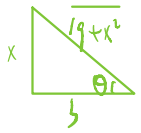


$x = a\sin\theta$

$\sin\theta = \frac{x}{a}$

$\theta = \arcsin\left(\frac{x}{a}\right)$

Example 7.20. Find $\int \frac{dx}{\sqrt{9+x^2}}$.



$dx = 3\sec^2\theta\, d\theta$

Let $x = 3\tan\theta$; $\frac{dx}{d\theta} = 3\sec^2\theta$

$\tan\theta = \frac{x}{3}$

$\theta = \arctan\left(\frac{x}{3}\right)$

$\cos\theta = \frac{3}{\sqrt{9+x^2}}$

$= \int \frac{1}{\sqrt{9+9\tan^2\theta}}\, dx$

$= \int \frac{1}{3\sqrt{1+\tan^2\theta}}\,(3\sec^2\theta)\, d\theta$

$= \int \frac{\sec^2\theta}{\sqrt{1+\tan^2\theta}}\, dx \qquad 1+\tan^2 x = \sec^2 x$

$= \int \frac{\sec^2\theta}{\sec\theta}\, dx$

$= \int \sec\theta\, dx$

$= \int \sec\theta \cdot \frac{\sec\theta+\tan\theta}{\sec\theta+\tan\theta}\, d\theta \qquad = \ln|\sec\theta+\tan\theta| + C$

$= \ln\left|\frac{\sqrt{9+x^2}}{3} + \frac{x}{3}\right| + C$

$\tan\theta \quad \sec^2\theta$

Example 7.21. Find $\int_2^4 \frac{\sqrt{x^2-4}}{x}dx$.

Let $x = 2\sec\theta \Rightarrow \frac{dx}{d\theta} = 2\sec\theta\tan\theta$

$dx = 2\sec\theta\tan\theta\, d\theta$

$\theta = \sec^{-1}\left(\frac{x}{2}\right)$

When $x=4$, $\theta = \sec^{-1}\left(\frac{4}{2}\right) = \sec^{-1}(2) = \frac{\pi}{3}$

$x=2$, $\theta = \sec^{-1}\left(\frac{2}{2}\right) = \sec^{-1}(1) = 0$

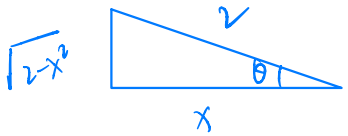
$\int_0^{\frac{\pi}{3}} \frac{\sqrt{4\sec^2\theta-4}}{2\sec\theta} \cdot 2\sec\theta\tan\theta\, d\theta$

$= \int_0^{\frac{\pi}{3}} 2\tan\theta \cdot \tan\theta\, d\theta$

$= 2\int_0^{\frac{\pi}{3}} \sec^2\theta - 1\, d\theta = 2\left[\tan\theta - \theta\right]_0^{\frac{\pi}{3}}$

$= 2\left[\tan\frac{\pi}{3} - \frac{\pi}{3} - (\tan 0 - 0)\right]$

$= 1.3697 \Leftrightarrow 2\left(\sqrt{3} - \frac{\pi}{3}\right)$



$\tan\theta = \frac{\sqrt{2-x^2}}{x}$

Example 7.22. Find $\displaystyle\int \frac{dx}{\sqrt{x^2+2x+26}}$.

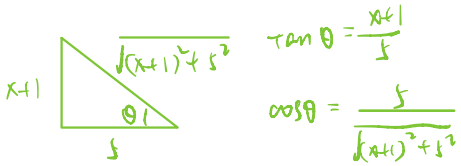
Expression	Substitution	Domain	Identity
$\sqrt{a^2-x^2}$	$x=a\sin\theta$	$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	$1-\sin^2\theta=\cos^2\theta$
$\sqrt{a^2+x^2}$	$x=a\tan\theta$	$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	$1+\tan^2\theta=\sec^2\theta$
$\sqrt{x^2-a^2}$	$x=a\sec\theta$	$0\le\theta\le\pi,\ \theta\ne\frac{\pi}{2}$	$\sec^2\theta-1=\tan^2\theta$

$= \int \frac{dx}{\sqrt{x^2+2x+1^2+26-1^2}}$

$= \int \frac{dx}{\sqrt{(x+1)^2+25}}$ Change to similar shape

let $x+1 = 5\tan\theta \Rightarrow \frac{dx+1}{d\theta} = 5\sec^2\theta \Rightarrow dx = 5\sec^2\theta\, d\theta$

$= \int \frac{5\sec^2\theta\, d\theta}{\sqrt{(5\tan\theta)^2+25}}$



$= \int \frac{5\sec^2\theta\, d\theta}{5\sqrt{\sec^2\theta}} = \int \sec\theta\, d\theta$

$= \ln|\sec\theta+\tan\theta| + C$

$= \ln\left|\frac{\sqrt{(x+1)^2+5^2}+x+1}{5}\right| + C$

Example 7.23. Find $\displaystyle\int \frac{1}{x^2\sqrt{x^2+4}}dx$.

let $x = 2\tan\theta$; $dx = 2\sec^2\theta\, d\theta$

$= \int \frac{1}{4\tan^2\theta\sqrt{4\tan^2\theta+4}} \cdot 2\sec^2\theta\, d\theta$

$= \int \frac{\sec\theta}{4\tan^2\theta}\, d\theta$

$= \frac{1}{4}\int \frac{\frac{1}{\cos\theta}}{\frac{\sin^2\theta}{\cos^2\theta}}\, d\theta$

$= \frac{1}{4}\int \frac{\cos\theta}{\sin^2\theta}\, d\theta$

let $u = \sin\theta$; $\frac{du}{d\theta} = \cos\theta$; $d\theta = \frac{1}{\cos\theta}\, du$

$= \frac{1}{4}\int \frac{1}{u^2}\, du$

$= -\frac{1}{4}u^{-1} + C$

$= -\frac{1}{4}\times\frac{1}{2\tan\theta} + C = -\frac{1}{2\tan\theta} + C$

Example 7.24. Find $\int \frac{1}{\sqrt{x^2-a^2}}dx$.

$\cos^2\theta + \sin^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$\sec^2\theta - 1 = \tan^2\theta$

Let $x = a\sec\theta$; $dx = a\sec\theta\tan\theta\, d\theta$

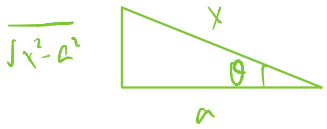
$= \int \frac{1}{\sqrt{a^2\sec^2\theta - a^2}} \cdot a\sec\theta\tan\theta\, d\theta$

$= \int \sec\theta\, d\theta$

$= \ln|\sec\theta + \tan\theta| + C$

$= \ln\left|\frac{x}{a} + \frac{\sqrt{x^2-a^2}}{a}\right| + C$

$\tan\theta = \frac{\sqrt{x^2-a^2}}{a}$



Example 7.25. Find $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2+9)^{3/2}}dx$.

Let $2x = 3\tan\theta$

$x = \frac{3}{2}\tan\theta$; $dx = \frac{3}{2}\sec^2\theta\, d\theta$

$\theta = \tan^{-1}\left(\frac{2x}{3}\right)$; $x = 0 \Rightarrow \theta = \tan^{-1}(0) = 0$

$x = \frac{3\sqrt{3}}{2} \Rightarrow \theta = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

$= \int_0^{\frac{\pi}{3}} \frac{\left(\frac{3}{2}\tan\theta\right)^3}{\left(9\tan^2\theta + 9\right)^{\frac{3}{2}}} \cdot \frac{3}{2}\sec^2\theta\, d\theta$

$= \frac{3}{2}\int_0^{\frac{\pi}{3}} \frac{\frac{27}{8}\tan^3\theta}{27\sec^3\theta} \cdot \sec^2\theta\, d\theta$

$= \frac{3}{16}\int_0^{\frac{\pi}{3}} \frac{\tan^3\theta}{\sec\theta}\, d\theta$

$= \frac{3}{16}\int_0^{\frac{\pi}{3}} \frac{\sin^3\theta}{\cos^3\theta} \cdot \cos\theta\, d\theta$

$= \frac{3}{16}\int_0^{\frac{\pi}{3}} \frac{\sin^2\theta}{\cos^2\theta}\sin\theta\, d\theta = \frac{3}{16}\int_0^{\frac{\pi}{3}} \frac{1-\cos^2\theta}{\cos^2\theta}\sin\theta\, d\theta$

Let $u = \cos\theta$; $\frac{du}{d\theta} = -\sin\theta$

$d\theta = \frac{du}{-\sin\theta}$

$\theta = \frac{\pi}{3}$, $u = \frac{1}{2}$; $\theta = 0$, $u = 1$

$= -\frac{3}{16}\int_1^{\frac{1}{2}} \frac{1-u^2}{u^2}\, du$

$= \frac{3}{16}\int_{\frac{1}{2}}^1 u^{-2} - 1\, du$

$= \frac{3}{16}\left[-u^{-1} - u\right]_{\frac{1}{2}}^1$

$= \frac{3}{16}\left[(-1-1) - \left(-2 - \frac{1}{2}\right)\right]$

$= \frac{1}{2} \times \frac{3}{16} = \frac{3}{32}$

Example 7.26. Find $\int \dfrac{x}{\sqrt{3-2x-x^2}}dx$.

$$= \int \frac{x}{\sqrt{-(x+1)^2+4}}\,dx$$

$$= \int \frac{x}{\sqrt{4-(x+1)^2}}\,dx$$

let $x+1 = 2\sin\theta \Rightarrow x = 2\sin\theta - 1 \Rightarrow dx = 2\cos\theta\,d\theta$

$$= \int \frac{2\sin\theta - 1}{\sqrt{4-4\sin^2\theta}} \cdot 2\cos\theta\,d\theta$$

$$= \int \frac{2\sin\theta - 1}{2\cos\theta} \cdot 2\cos\theta\,d\theta$$

$$= \int 2\sin\theta - 1\,d\theta$$

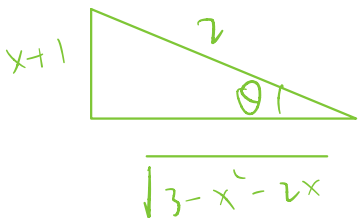
$$= -2\cos\theta - \theta + C$$

$$= -\sqrt{3-x^2-2x} - \sin^{-1}\left(\frac{x+1}{2}\right) + C$$

$\sin\theta = \dfrac{x+1}{2}$

$\theta = \sin^{-1}\left(\dfrac{x+1}{2}\right)$

$\cos\theta = \dfrac{\sqrt{3-x^2-2x}}{2}$



Exercises 7.3 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 23, 25, 27, 29, 37.

I. 分子 degree < 分母 degree

Case I：有兩個不同的

$$\int \frac{\quad}{\quad} dx = \int \frac{A}{\quad} + \frac{B}{\quad} dx$$

. Find $\int \frac{5x+3}{x^3-2x^2-3x}dx.$

$= \int \frac{5x+3}{x(x+1)(x-3)} dx$

$= \int \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x+1} dx$

Case II. 有 2 個相同的

(1) . Find $\int \frac{x}{(x-3)^2}dx.$

$= \int \frac{A}{x-3} + \frac{B}{(x-3)^2} dx$

(2) 有 1 個不同，2 個相同的

Find $\int \frac{3x^2-8x+13}{(x+3)(x-1)^2}dx.$

$= \int \frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{(x-1)^2} dx$

Case III 有 complex 的

5. Find $\int \frac{6x^2-3x+1}{(4x+1)(x^2+1)}dx.$

$= \int \frac{A}{4x+1} + \frac{Bx+C}{x^2+1} dx$

III 分子 degree = 分母 degree

相近 => similar case

$$= \int \frac{2u^2}{u^2-4}\,du = \int \frac{2u^2-8+8}{u^2-4}\,du$$

次数相同

$$= \int \frac{2(u^2-4)+8}{u^2-4}\,du$$

$$= \int 2 + \frac{8}{u^2-4}\,du$$

$$= 2u + 8\int \frac{1}{u^2-4}\,du$$

$$= 2u + 8\int \frac{1}{(u+2)(u-2)}\,du$$

$$= 2u + 8\int \frac{A}{u+2} + \frac{B}{u-2}\,du$$

$$= 2u + 8\int \cdot\frac{1}{4}\cdot\frac{1}{u+2} + \frac{1}{4}\frac{1}{u-2}\,du$$

$$= 2u - 2\ln|u+2| + 2\ln|u-2| + C$$

$$= 2\sqrt{x+4} - 2\ln|\sqrt{x+4}+2| + 2\ln|\sqrt{x+4}-2| + C$$

$1 = A(u-2) + B(u+2)$

$u = 2$ $\quad 1 = 4B$ $\quad B = \frac{1}{4}$

$u = -2$ $\quad 1 = -4A$ $\quad A = -\frac{1}{4}$

分子 degree > 分母 degree

Long division

7.4 Integration of Rational Functions Using Partial Fractions

Example 7.27. Find $\int \frac{2}{(x+1)^3}dx.$

eg. 7.7

Example 7.28. Find $\int \frac{x^3+x}{x-1}dx.$

Improper rational function

分母的 degree < 分子的 degree

Apply long division

$$= \int x^2+x+2+\frac{2}{x-1}\,dx$$

$$= \frac{x^3}{3}+\frac{x^2}{2}+2x+2\ln|x-1|+C \quad \text{let } u=x-1$$

$$\begin{array}{r} x^2+x+2 \\ x-1 \overline{)\,x^3+0x^2+x+0} \\ \underline{-x^3+x^2\ \ \ } \\ x^2+x \\ \underline{-x^2+x\ \ } \\ 2x+0 \\ \underline{-2x+2} \\ 2 \end{array}$$

Example 7.29. Find $\int \frac{2x+2}{x^2-4x+8}dx.$

degree 1 / degree 2

completing the square

$$= \int \frac{2x+2}{(x-2)^2+4}\,dx$$

let $u=x-2$; $\frac{du}{dx}=1 \Rightarrow dx=du$

$x=u+2$

$$= \int \frac{2u+6}{u^2+4}\,du$$

$$= \int \frac{2u}{u^2+4}+\frac{6}{u^2+4}\,du \quad \text{substitution}$$

let $w=u^2+4 \Rightarrow \frac{dw}{du}=2u \Rightarrow du=\frac{dw}{2u}$

$$= \ln|u^2+4| + 6\left[\tfrac{1}{2}\tan^{-1}\left(\tfrac{u}{2}\right)\right]+C \qquad = \ln|(x-2)^2+4| + 3\tan^{-1}\left(\tfrac{x-2}{2}\right)+C$$

- $D_x \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$ (inverse of sin)
- $D_x \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$
- $D_x \tan^{-1} x = \frac{1}{1+x^2}$

13. $\int \frac{1}{1+u^2}du = \tan^{-1} u + C$

$\int \frac{1}{a+u^2}\,du = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right)+C$

Case I Distinct roots with degree of $\square$ < degree of $\square$

Example 7.30. Decompose $\dfrac{3x-1}{x^2-x-6}$ into partial fractions and find $\displaystyle\int \frac{3x-1}{x^2-x-6}dx$.

$$= \frac{3x-1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$$

$$= \frac{A(x+2)+B(x-3)}{(x-3)(x+2)}$$

Method 1: ① let $x=$ one of the roots $\begin{cases} x+2=0 \Rightarrow x=-2 \\ x-3=0 \Rightarrow x=3 \end{cases}$

let $x=3$: $8=5A$, $A=\frac{8}{5}$

let $x=-2$: $-7=-5B$, $B=\frac{7}{5}$

代入 ① $3x-1 = A(x+2)+B(x-3)$

Method 2

$(A+B)x + 2A-3B = 3x-1$

$\begin{cases} A+B=3 \\ 2A-3B=-1 \end{cases}$

$$= \int \frac{\frac{8}{5}}{x-3} + \frac{\frac{7}{5}}{x+2}\,dx$$

$$= \frac{8}{5}\ln|x-3| + \frac{7}{5}\ln|x+2| + C$$

Case I

Example 7.31. Find $\displaystyle\int \frac{5x+3}{x^3-2x^2-3x}dx.$ $A\ln|x| + B\ln|x-3| + C\ln|x-3|$ ($A=-1$, $B=\frac{3}{2}$, $C=-\frac{1}{2}$)

$$= \int \frac{5x+3}{x(x+1)(x-3)}\,dx$$

$$= \int \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x+1}\,dx$$

$$= \int \frac{A(x-3)(x+1) + Bx(x+1) + Cx(x-3)}{x(x-3)(x+1)}\,dx$$

$$5x+3 = A(x-3)(x+1) + Bx(x+1) + Cx(x-3)$$

① $x=3$

$18 = 12B \Rightarrow B = \frac{3}{2}$

② $x=-1$

$-2 = 4C \Rightarrow C = -\frac{1}{2}$

③ $x=0$

$3 = -3A$

$A = -1$

$$= \int -\frac{1}{x} + \frac{3}{2}\cdot\frac{1}{x-3} - \frac{1}{2}\cdot\frac{1}{x+1}\,dx$$

$$= -\ln|x| + \frac{3}{2}\ln|x-3| - \frac{1}{2}\ln|x+1| + C$$

Case II: repeated roots

Example 7.32. Find $\int \frac{x}{(x-3)^2}dx$.

$= \int \frac{A}{x-3} + \frac{B}{(x-3)^2}\,dx$ (if the roots are distinct)

$= \int \frac{A}{x-3} + \frac{B}{(x-3)^2}\,dx$

$= \int \frac{1}{x-3} + \frac{3}{(x-3)^2}\,dx$

$\int \frac{x}{(x-3)^2}dx = \int \frac{A}{(x-3)^{n1}} \frac{B}{(x-3)^2}\,dx$

let $u = x-3$; $\frac{du}{dx} = 1$

$= \ln|x-3| - \frac{3}{x-3} + C$

$x = A(x-3) + B$

$x = A(x-3) + B$

let $x = 3$ | let $x = 0$

$3 = B$ | $0 = -3A + 3$

$A = 1$

Try

Example 7.33. Find $\int \dfrac{3x^2-8x+13}{(x+3)(x-1)^2}dx$.

$$= \int \frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{(x-1)^2}\,dx$$

$$3x^2-8x+13 = A(x-1)^2 + B(x+3)(x-1) + C(x+3)$$

① let $x=-3$

$64 = 16A$

$64 = -64A \Rightarrow A = -1$ ✗ $A = 4$

② let $x=1$

$8 = 4C \Rightarrow C = 2$ ✓

$8 = 4C \Rightarrow C = 2$

③ let $x=0$

$13 = 1 + 3B - 3C$

$B = 6$ ✗

$$= \int -\frac{1}{x+3} + 6\cdot\frac{1}{x-1} + 2\frac{1}{(x-1)^2}\,dx$$

let $u = x-1$; $du = dx$

$$= -\ln|x+3| + 6\ln|x-1| + 2\int \frac{1}{u^2}\,du$$

$$- - - -

Example 7.34. Find $\displaystyle\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx.$

degree ∨ degree

$$x^3 - x^2 - x + 1 \;\overline{)\; x^4 + 0x^3 - 2x^2 + 4x + 1}$$

$$x + 1$$

$$x^4 - x^3 - x^2 + x$$

$$0 + x^3 - x^2 + 3x + 1$$

$$x^3 - x^2 - x + 1$$

$$0 \quad 0 \quad 4x + 0$$

$(x-1)$

$(x+1)$

$x = 1$

$x = -1$

$$= \int x + 1 + \frac{4x}{x^3 - x^2 - x + 1}\, dx$$

↓ 각 항을 분수로 만들기 위해 곱하기

$$= \int x + 1 + \frac{4x}{(x+1)(x-1)^2}\, dx$$

$$x+1 \;\overline{)\; x^3 - x^2 - x + 1}$$

$$x^2 - 2x + 1$$

$$x^3 + x^2$$

$$0 \quad -2x^2 - x$$

$$-2x^2 - 2x$$

$$0 \quad +x + 1$$

$$x + 1$$

$$0$$

$$= \frac{x^2}{2} + x + \int \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}\, dx$$

$$= \frac{x^2}{2} + x + \int -\frac{1}{x+1} - \frac{1}{x-1} + \frac{2}{(x-1)^2}\, dx$$

$$= \frac{x^2}{2} + x - \ln|x+1| - \ln|x-1| - \frac{2}{x-1} + C$$

$4x = A(x-1)^2 + B(x+1)(x-1) + C(x+1)$

$4x = A(x-1)^2 + B(x^2-1) + C(x+1)$

① $x = 1$

$4 = 2C \Rightarrow C = 2$

② $x = -1$

$-4 = 4A \Rightarrow A = -1$

③ $x = 0$

$0 = A + B + C$

$B = -1$

$u = x - 1$

2

Case II : complex 1-root

Example 7.35. Find $\int \dfrac{6x^2-3x+1}{(4x+1)(x^2+1)}dx$.

$$= \int \frac{A}{4x+1} + \frac{Bx+C}{x^2+1}\,dx$$

$$6x^2-3x+1 = A(x^2+1)+(Bx+C)(4x+1)$$

① when $x=-\frac{1}{4}$

$$\frac{17}{8} = \frac{17}{16}A \Rightarrow A=2$$

② when $x=0$

$$1 = A+C$$

$$1 = 2+C \Rightarrow C=-1$$

③ when $x=1$

$$4 = 2A+5B+5C$$

$$B=1$$

let $u = x^2+1$; $\frac{du}{dx} = 2x$

$dx = \frac{du}{2x}$

$$= \int \frac{2}{4x+1} + \frac{x-1}{x^2+1}\,dx$$

$$= \int \frac{1}{4x+1} + \frac{x}{x^2+1} - \frac{1}{x^2+1}\,dx$$

let $u = x^2+1$; $dx = \frac{du}{2x}$

$$= \frac{1}{4}\ln|4x+1| + \frac{1}{2}\ln|x^2+1| - \tan^{-1}(x) + C$$

Example 7.36. Write out the form of the partial fraction decomposition of the function

$$\frac{x^3+x^2+1}{x(x-1)(x^2+x+1)(x^2+1)^3}$$

Example 7.37. Evaluate $\int \frac{\sqrt{x+4}}{x}dx.$

let $u=\sqrt{x+4}$

$x=u^2-4 \;;\; dx=2u\,du$

$$=\int \frac{u}{u^2-4}\cdot 2u\,du$$

$$=\int \frac{2u^2}{u^2-4}\,du = \int \frac{2u^2-8+8}{u^2-4}\,du$$

$$=\int \frac{2(u^2-4)+8}{u^2-4}\,du$$

$$=\int 2+\frac{8}{u^2-4}\,du$$

$$=2u+8\int \frac{1}{u^2-4}\,du$$

$$=2u+8\int \frac{1}{(u+2)(u-2)}\,du$$

$$=2u+8\int \frac{A}{u+2}+\frac{B}{u-2}\,du$$

$$=2u+8\int \frac{1}{4}\cdot\frac{1}{u+2}+\frac{1}{4}\frac{1}{u-2}\,du$$

$$=2u-2\ln|u+2|+2\ln|u-2|+C$$

$$=2\sqrt{x+4}-2\ln|\sqrt{x+4}+2|+2\ln|\sqrt{x+4}-2|+C$$

$1=A(u-2)+B(u+2)$

$u=2 \qquad u=-2$

$1=4B \qquad 1=-4A$

$B=\frac{1}{4} \qquad A=-\frac{1}{4}$

Exercises 7.4 1, 3, 5, 7, 11, 15, 17, 19, 21, 23, 27, 29, 31, 33, 35, 37, 39, 41, 47, 49, 51, 65.

7.5 Strategies for Integration

- **Standard Integral Formulas**

1. $\int u^r du = \dfrac{u^{r+1}}{r+1} \quad (r \neq -1)$
2. $\int \dfrac{1}{u} du = \ln|u| + C$
3. $\int e^u du = e^u + C$
4. $\int a^u du = \dfrac{a^u}{\ln a} + C \quad (a > 0,\ a \neq -1)$
5. $\int \sin u du = -\cos u + C$
6. $\int \cos u du = \sin u + C$
7. $\int \sec^2 u du = \tan u + C$
8. $\int \csc^2 u du = -\cot u + C$
9. $\int \sec u \tan u du = \sec u + C$
10. $\int \csc u \cot u du = -\csc u + C$
11. $\int \tan u du = \ln|\sec u| + C$
12. $\int \cot u du = \ln|\sin u| + C$
13. $\int \sec u du = \ln|\sec u + \tan u| + C$
14. $\int \csc u du = -\ln|\csc u + \cot u| + C$
15. $\int \dfrac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1}\dfrac{u}{a} + C$
16. $\int \dfrac{1}{a^2 + u^2} du = \dfrac{1}{a}\tan^{-1}\dfrac{u}{a} + C$
17. $\int \sinh u du = \cosh u + C$
18. $\int \cosh u du = \sinh u + C$

Example 7.38. Evaluate $\int \frac{\tan^3 x}{\cos^3 x} dx$.

เก็บ $\tan x$ และ $\sec x$

$\cos^2 x + \sin^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$= \int \tan^3 x \sec^3 x \, dx$

$= \int \sec^2 x \cdot \tan^2 x \cdot \sec x \tan x \, dx$

let $u = \sec x$; $\frac{du}{dx} = \sec x \tan x$ $\Rightarrow$ $dx = \frac{du}{\sec x \tan x}$

$= \int \sec^2 x (\sec^2 x - 1) \sec x \tan x \, dx$

$= \int u^2 (u^2 - 1) \, du$

$= \int u^4 - u^2 \, du$

$= \frac{u^5}{5} - \frac{u^3}{3} + C$

$= \frac{1}{5} \sec^5 x - \frac{1}{3} \sec^3 x + C$

Example 7.39. Evaluate $\int e^{\sqrt{x}} dx$.

let $u = \sqrt{x}$; $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$; $dx = 2\sqrt{x} \, du$

$x = u^2$ $= 2u \, du$

$= \int e^u \cdot 2u \, du$ (integration by parts)

let $w = u$, $dv = e^u du$

$dw = du$ $v = e^u$

$= 2ue^u - 2\int e^u \, du$

$= 2ue^u - 2e^u + C$

$= 2\sqrt{x} e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$

Example 7.40. Evaluate $\int \frac{dx}{x\sqrt{\ln x}}$.

let $u = \ln x$; $\frac{du}{dx} = \frac{1}{x}$ $\Rightarrow dx = x\,du$

$$\int \frac{1}{x\sqrt{\ln x}} \cdot x\,du = \int \frac{1}{\sqrt{u}}\,du \qquad u^{-\frac{1}{2}}, \quad 2u^{\frac{1}{2}}$$

$$= 2u^{\frac{1}{2}} + C$$

$$= 2\sqrt{\ln x} + C$$

Example 7.41. Evaluate $\int \sqrt{\frac{1-x}{1+x}}\,dx$.

let $x = \cos 2\theta$; $dx = -2\sin 2\theta\,d\theta = -4\sin\theta\cos\theta\,d\theta$

$$= \int \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}}\,(-4\sin\theta\cos\theta)\,d\theta$$

$$= \int \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}}\,(\cdots)\,d\theta$$

$$= \int \tan\theta\,(-4\sin\theta\cos\theta)\,d\theta = -4\int \sin^2\theta\,d\theta$$

$$= -4\int \frac{1-\cos 2\theta}{2}\,d\theta$$

Exercises 7.5 1, 7, 19, 21, 31, 45, 57, 61, 63, 71, 75.

7.8 Improper Integrals

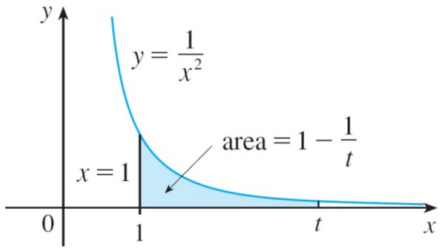
Infinite Intervals

- Integrals of the form

$$\int_0^{\infty} \frac{1}{1+x^2}dx, \quad \int_{-\infty}^{-1} xe^{-x^2}dx \quad \int_{-\infty}^{\infty} x^2e^{-x^2}dx$$

are called **improper integrals** with infinite limits.

- Consider $\int_1^{\infty} \frac{1}{x^2}dx$.



$$\int_1^{\infty} \frac{1}{x^2}dx = \lim_{t\to\infty}\int_1^t x^{-2}dx$$

$$= \lim_{t\to\infty}\left[-\frac{1}{x}\right]_1^t$$

$$= \lim_{t\to\infty} -\frac{1}{t} + 1$$

$$= 1$$

- **Definition**

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx,$$

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$$

If the limits on the right exists and have finite values, we say that the corresponding improper integrals **converge** and have those values. Otherwise, we say that the improper integrals **diverge**.

substitution (換 u)
integration by parts (uv)
三角 (trigo integral)
換元 三角

Example 7.42. Find, if possible, $\int_{-\infty}^{-1} xe^{-x^2}dx$.

let $u = -x^2$, $\frac{du}{dx} = -2x$; $dx = \frac{du}{-2x}$

when $x = -1$, $u = -1$

$x \to -\infty$, $u \to -\infty$

$$\int_{-\infty}^{-1} x e^{-x^2} dx = \lim_{t\to-\infty} \int_{t}^{-1} x e^{-u} \cdot \frac{du}{-2x}$$

$$= -\frac{1}{2} \lim_{t\to-\infty} \int_{t}^{-1} e^{+u} du$$

$$= -\frac{1}{2} \lim_{t\to-\infty} [+e^{+u}]_{t}^{-1}$$

$$= -\frac{1}{2} \lim_{t\to-\infty} (+e^{-1} - 0)$$

$$= \frac{0}{2} - \frac{1}{2e}$$

Example 7.43. Find, if possible, $\int_0^{\infty} \sin x dx$.

$$= \lim_{t\to\infty} \int_0^t \sin x \, dx$$

$$= \lim_{t\to\infty} [-\cos x]_0^t$$

$$= \lim_{t\to\infty} (-\cos t + 1)$$

$\int_0^{\infty} \sin x dx$ is divergent

Try

Example 7.44. For what values of p does the integral $\int_1^\infty \frac{1}{x^p}dx$ converge?

$p > 1$, converges

Both Limits Infinite

- **Definition**

If both $\int_{-\infty}^{a} f(x)dx$ and $\int_{a}^{\infty} f(x)dx$ converge, then $\int_{-\infty}^{\infty} f(x)dx$ is said to converge and have value

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{a} f(x)dx + \int_{a}^{\infty} f(x)dx$$

Otherwise, $\int_{-\infty}^{\infty} f(x)dx$ diverges.

Example 7.45. Determine whether $\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx$ is convergent. If yes, find its value.

步一个好算的 c within 定义域)

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx$$

$$= \lim_{t\to-\infty} \int_{t}^{0} \frac{1}{1+x^2}\,dx + \lim_{s\to\infty} \int_{0}^{s} \frac{1}{1+x^2}\,dx$$

$$= \lim_{t\to-\infty} \left[-\tan^{-1}x\right]_{t}^{0} + \lim_{s\to\infty} \left[\tan^{-1}x\right]_{0}^{s}$$

$$= \lim_{t\to-\infty} \left(0 - \tan^{-1}t\right) + \lim_{s\to\infty} \left(\tan^{-1}s - \tan^{-1}0\right)$$

$$= \pi$$

Try Ans : 0

Example 7.46. Determine whether $\int_{-\infty}^{\infty} \frac{x}{1+x^2} dx$ is convergent. If yes, find its value.

Discontinuous Integrands

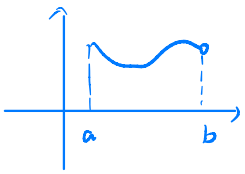
Example 7.47. Evaluate $\displaystyle\int_{-2}^{1} \frac{1}{x^2}dx.$

Integrands Infinite at an End Point

- **Definition**

Let f be continuous on the half-open interval $[a, b)$ and suppose that $\lim_{x\to b^-} |f(x)| = \infty$, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$$



provided that this limit exists and is finite, in which case we say that the integral converges. Otherwise, we say that the integral diverges.

Let f be continuous on the half-open interval $(a, b]$ and suppose that $\lim_{x\to a^+} |f(x)| = \infty$, then

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$$

provided that this limit exists and is finite, in which case we say that the integral converges. Otherwise, we say that the integral diverges.

Example 7.48. Evaluate, if possible, the improper integral $\int_0^2 \frac{dx}{\sqrt{4-x^2}}$.

$x \neq 2$

$$\int_0^2 \frac{dx}{\sqrt{4-x^2}} = \lim_{t\to 2^-} \int_0^t \frac{dx}{\sqrt{4-x^2}}$$

$\left(\because \frac{\pi}{2} < \theta < \frac{\pi}{2}\right)$

$\theta = \sin^{-1}\left(\frac{x}{2}\right)$

let $x = 2\sin\theta$; $\frac{dx}{d\theta} = 2\cos\theta \Rightarrow d\theta = \frac{dx}{2\cos\theta}$

$$= \lim_{t\to 0^-} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{4-4\sin^2\theta}} \cdot 2\cos\theta \cdot d\theta$$

when $x=0$, $\theta \to 0$

$x \to 2$, $\theta \to \frac{\pi}{2}$

$$= \lim_{t\to 0^-} \int_0^{\frac{\pi}{2}} 1 \, d\theta$$

$$= \lim_{t\to 0^-} [\theta]_0^{\frac{\pi}{2}}$$

$\lim_{t\to 1^-} \left[\sin^{-1}\left(\frac{x}{2}\right)\right]_0^t$

$$= \frac{\pi}{2}$$

Example 7.49. Evaluate, if possible, $\int_0^{16} \frac{1}{\sqrt[4]{x}} dx$.

$$= \lim_{t\to 0^+} \int_t^{16} \frac{1}{\sqrt[4]{x}} \, dx$$

$$= \frac{4}{3} \lim_{t\to 0^+} \left[x^{\frac{3}{4}}\right]_t^{16}$$

$$= \frac{4}{3} \lim_{t\to 0^+} \left(16^{\frac{3}{4}} - t^{\frac{3}{4}}\right)$$

$$= \frac{4}{3} \times 16^{\frac{3}{4}}$$

$-\frac{1}{4}$

x

$\frac{4}{3} \quad x^{\frac{3}{4}}$

$-\frac{1}{4}$

x

$\frac{4}{3} \lim_{t\to 0^+} \left[x^{\frac{3}{4}}\right]_t^{16}$

$\lim_{t\to 0^+} \int_t^{16} \frac{1}{\sqrt[4]{x}} \, dx$

Example 7.50. Evaluate, if possible, $\int_0^1 \frac{1}{x} dx$.

$= \lim_{t \to 0^+} \int_t^1 \frac{1}{x} dx$

$= \lim_{t \to 0^+} [\ln|x|]_t^1$

$= \lim_{t \to 0^+} (0 - \ln|t|)$

$= -\infty$

$\int_0^1 \frac{1}{x} dx$ is divergent

$\lim_{t \to 0^+} \int_t^1 \frac{1}{x} dx$

$\lim_{t \to 0^+} [\ln|x|]_t^1$

Example 7.51. For what values of p does the integral $\int_0^1 \frac{1}{x^p} dx$ converge?

LIATE

Example 7.52. Evaluate, if possible, $\int_0^1 \ln x dx$.

$$= \lim_{t \to 0^+} \int_t^1 \ln x \, dx$$

$$= \lim_{t \to 0^+} \left[x \ln x - x \right]_t^1$$

$$= (0 - 1) - \lim_{t \to 0^+} (t \ln t - t)$$

let $u = \ln x$; $dv = dx$

$du = \frac{1}{x} dx$, $v = x$

$$\lim_{t \to 0^+} \int_t^1 \ln x \, dx$$

$$= \lim_{t \to 0^+} \left[x \ln x - \int x \cdot \frac{1}{x} dx \right]_t^1$$

$$= \lim_{t \to 0^+} \left[x \ln x - x \right]_t^1$$

$$\lim_{t \to 0^+} t \ln t = \lim_{t \to 0^+} \frac{\ln t}{\frac{1}{t}}$$

$$\overset{H}{=} \lim_{t \to 0^+} \frac{\frac{1}{t}}{-\frac{1}{t^2}}$$

$$= \lim_{t \to 0^+} -t$$

$$= 0$$

Integrands Infinite at an Interior Point

- **Definition**

Let f be continuous on $[a, b]$ except at a number $c \in (a, b)$, and suppose that $\lim_{x \to c} |f(x)| = \infty$. Then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)dx$$

provided both integrals on the right converge. Otherwise, we say that $\int_a^b f(x)dx$ diverges.

Example 7.53. Show that $\int_{-2}^{1} \frac{1}{x^2}dx$ diverges.

$$= \int_{-2}^{0} \frac{1}{x^2}\,dx + \int_{0}^{1} \frac{1}{x^2}\,dx$$

$$\int_{-2}^{0} \frac{1}{x^2}\,dx = \lim_{t \to 0^-} \int_{-2}^{t} \frac{1}{x^2}\,dx$$

$$= \lim_{t \to 0^-} \left[-\frac{1}{x}\right]_{-2}^{t}$$

$$= \lim_{t \to 0^-} \left[-\frac{1}{t} + \frac{1}{2}\right]$$

$$= -\infty$$

$\int_{-2}^{1} \frac{1}{x^2}\,dx$ is divergent.

Example 7.54. Evaluate, if possible, the improper integral $\displaystyle\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$.

Substitution rules

let $u = x-1$; $\frac{du}{dx} = 1$, $dx = du$

$$\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}} = \int_0^1 \frac{dx}{(x-1)^{\frac{2}{3}}} + \int_1^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$$

$$\lim_{t\to 1^-} \int_0^t \frac{dx}{(x-1)^{\frac{2}{3}}} = 3 \lim_{t\to 1^-} \left[(x-1)^{\frac{1}{3}} \right]_0^t = 3 \lim_{t\to 1^-} [0 - (-1)] = 3$$

$$\lim_{t\to 1^+} \int_t^3 \frac{dx}{(x-1)^{\frac{2}{3}}} = 3 \lim_{t\to 1^+} \left[(x-1)^{\frac{1}{3}} \right]_t^3 = 3 \cdot \sqrt[3]{2}$$

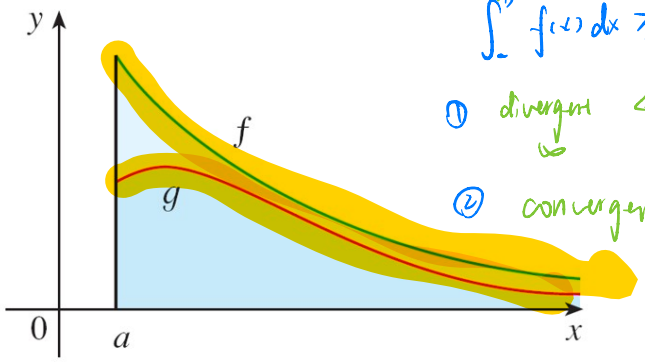
- **Comparison Test for Improper Integrals**

Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ when $x \geq a$.

integrable

(a) If $\int_a^\infty f(x)dx$ is convergent, so is $\int_a^\infty g(x)dx$.

(b) If $\int_a^\infty g(x)dx$ is divergent, so is $\int_a^\infty f(x)dx$.

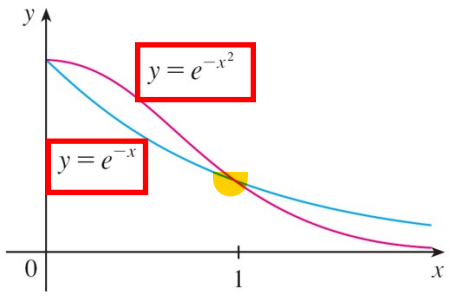


$\int_a^b f(x)dx \geq \int_a^b g(x)dx \geq 0$

① divergent ⇐ divergent

② convergent ⇒ convergent

Example 7.55. Determine whether $\int_0^\infty e^{-x^2}dx$ is convergent or divergent.



For $x > 1$, $e^{-x} > e^{-x^2}$

$\lim_{t\to\infty} \int_1^t e^{-x}dx = \lim_{t\to\infty} -[e^{-x}]_1^t = \lim_{t\to\infty} (-e^{-t} + e^{-1})$

$= e^{-1}$

$\therefore \int_0^\infty e^{-x^2}dx$ convergent

$\int_0^1 e^{-x^2}dx$ is a finite integral, which is convergent.

$\therefore \int_0^\infty e^{-x^2}dx$ is convergent

Example 7.56. Determine whether $\int_1^\infty \frac{1+e^{-x}}{x}dx$ is convergent or divergent.

$= \int_1^\infty \frac{1}{x} + \frac{e^{-x}}{x}\,dx$

$= \lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx + \lim_{t\to\infty} \int_1^t \frac{e^{-x}}{x}\,dx$

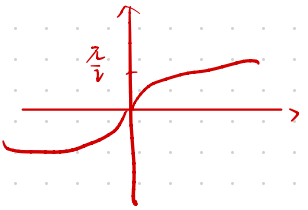
$\lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx = \lim_{t\to\infty} [\ln|x|]_1^t = \infty$ divergent

$\therefore \int_1^\infty \frac{1+e^{-x}}{x}\,dx$ divergent.

Exercises 7.8 5, 7, 9, 11, 13, 15, 17, 19, 23, 25, 31, 33, 37, 39, 49, 51, 53, 55, 59.

$$\int_0^{\infty} \frac{x}{x^3+1}\,dx$$

Another $\int_0^{\infty} \frac{1}{x^2+1}\,dx$



$$= \lim_{t\to\infty} \int_0^t \frac{1}{x^2+1}\,dx$$

$$= \lim_{t\to\infty} [\arctan x]_0^t$$

$$= \frac{\pi}{2} \quad \text{convergent}$$

$$\therefore \int_0^{\infty} \frac{x}{x^3+1}\,dx < \int_0^{\infty} \frac{1}{x^2+1}\,dx$$

$$\therefore \int_0^{\infty} \frac{x}{x^3+1}\,dx \text{ is convergent}$$